

preliminary

Inspired by BSI (2025) and Yakura et al. (2025) which found evidence for GPT influenced human language after the introduction of chatGPT we tried to replicate the Yakura et al. (2025) pipeline of building an AI vocabulary (model preferred lemmata) and compare frequencies of model typical words across pre- and post model introduction human language corpora. The first draft essai proves their hypothesis that LLM generated language manifests within human natural language.

```
# # tidyqu<-0  
library(quanteda)
```

questions / hypotheses

- do humans adapt to language produced by LLM and incorporate model vocabulary into their own language production?
- H1: we will find higher frequencies of LLM typical vocabulary within human natural language corpora after onset of model introduction.

```
library(lme4)  
f1h1$gus<-0
```

methods / data

Our human languaga data consists of raw texts from german bundestag plenary protocols, DIP (2026). The LLM corpus consists of model summaries of a first subset of these texts generated with the prompt (to gemini API, Wikipedia and Google (2026)) you see in C1. We first devised model typical lemmata in the AI corpus which are distinct for that corpus using their log odds score calculated over pre/post target, cf. Figure 3. For corpus building and evaluation scripts cf. Schwarz (2026).

```
# tu<-unique(tokens.2$word)
```

c1

System prompt: You are a member of german parliament. Prepare a summary of the text provided to present at a local community meeting of your party members. Output in german language, no preamble, no extra information, just the plain text. Wordcount maximal 300 words, containing not more than 5% of the keywords of the text provided and explicitly not just a list of keywords but an entertaining text. You are supposed to interpret freely, including background insights on daily politics. Keep in mind thatthe text will be used as is as keynotes to the talk being held to the locals. Text:

```
#
```

discussion / limitations

If the findings of increased AI lemma frequencies really prove manifestation of AI language depends also on influences of other factors of language variation and change which Yakura et al. (2025) already took into account. A deeper going analysis of general trends in language development and investigation on the bidirectional influences of model-training and human language could harden the results. Also dating of the adressed onset of gemini introduction and correspondingly synchronizing the target resp. the AI corpus date ranges is an issue not yet fully adressed within our pipeline. Since google gemini was introduced in different stages with changing model traing data, to exactly sync our corpora with the model output i.e. build the AI vocabulary corresponding with that timeframe is an open issue.

```
tab2 <-  
#mode()  
)  
get.dist.norm<-function(lmdf){  
head(lmdf)  
tok.r2<-tokens.r[tokens.r$word!="",]  
family = binomial,  
family = poisson(link = "log"))  
load(paste0(Sys.getenv("HKW_TOP"),"/SPUND/2025/huening/tok.na.ann.RData"))  
pcd<-pc8  
sum(f1h1$gus)
```

p2

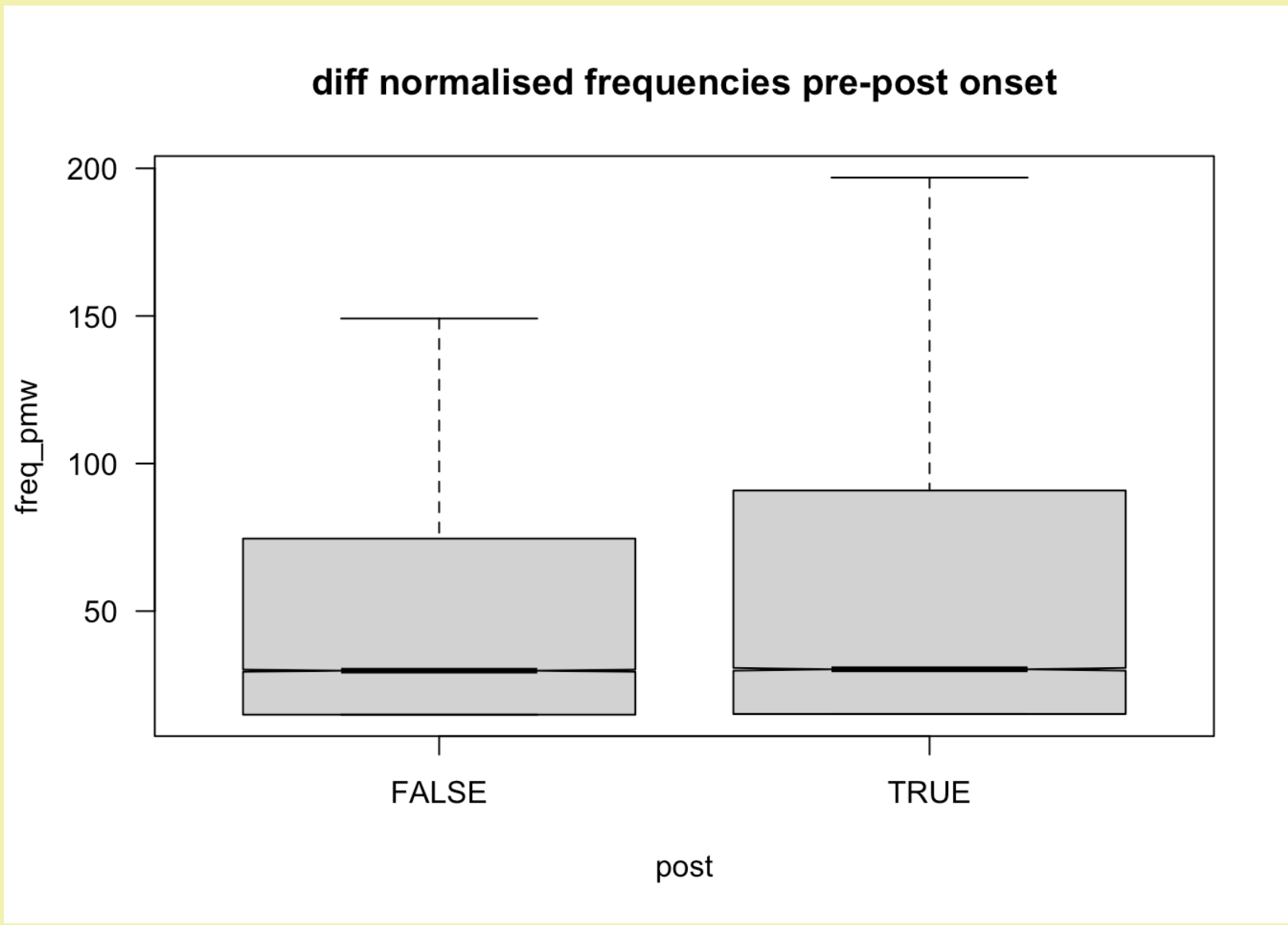


Figure 1: descriptive stats plot

p3

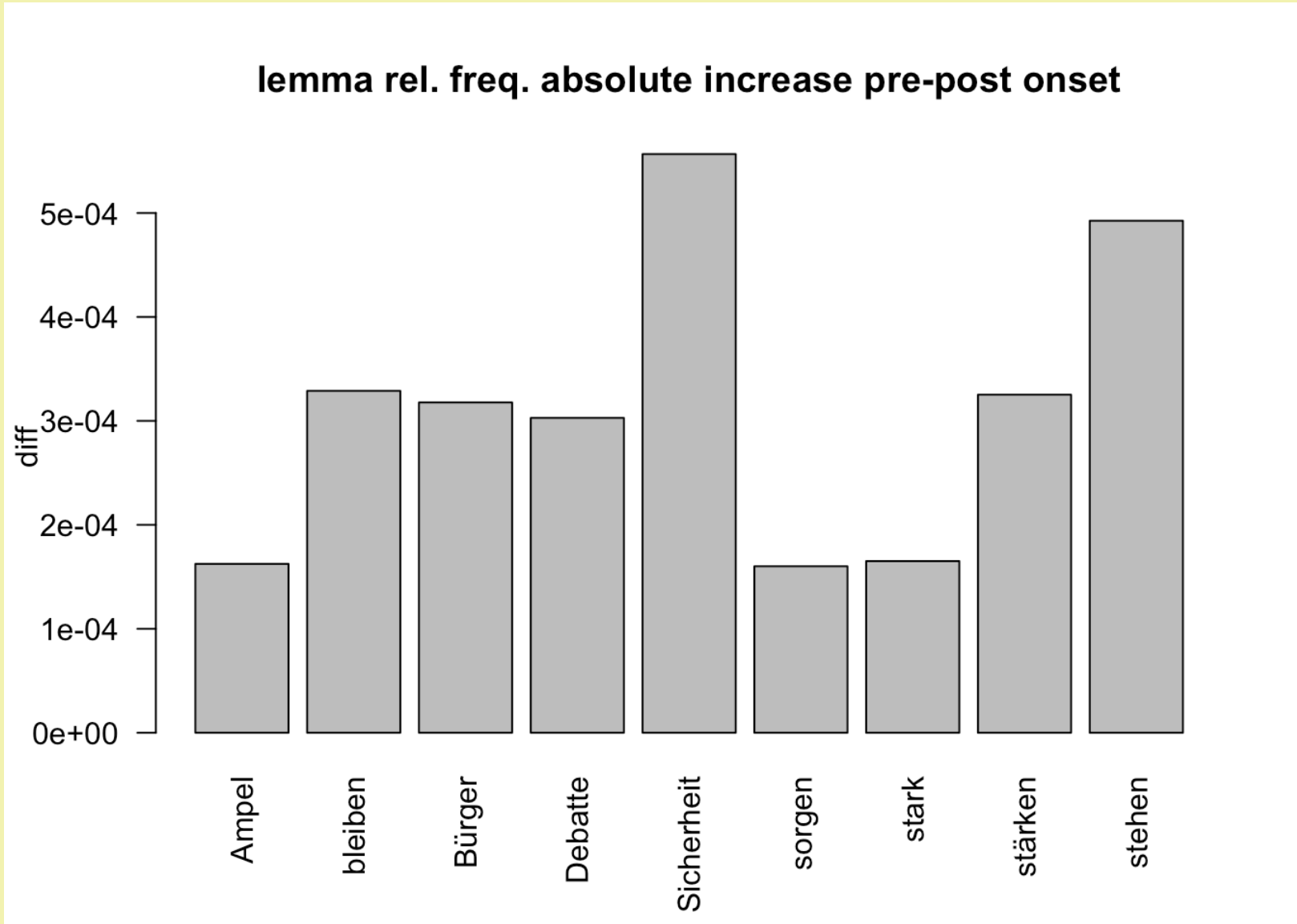


Figure 2: lemma frequency increase of relevant AI typical lemma

c2: corpus stats

stat	var	pre	post
mean	freq_rel gpt lemmas	0.000704	0.000751
mean	freq normalised pmw	29.825	30.288
tokens	n	1640446	1940385
statistics...			

c3: regression simple model

```
Call:  
lm(formula = freq_pmw ~ post + gp, data = df_lg.norm)  
  
Residuals:  
    Min       1Q   Median       3Q      Max  
-1400    -435    -388    -243   436922  
  
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)  
(Intercept)   833.75      38.64   21.576 < 2e-16 ***  
postTRUE       83.94      20.64    4.067 4.77e-05 ***  
gp             68.11       5.20    13.097 < 2e-16 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 3765 on 133087 degrees of freedom  
(1393 observations deleted due to missingness)  
Multiple R-squared:  0.0014,    Adjusted R-squared:  0.001385  
F-statistic: 93.32 on 2 and 133087 DF,  p-value: < 2.2e-16  
  
Analysis of Variance Table  
  
Response: freq_pmw  
              Df Sum Sq Mean Sq F value    Pr(>F)      
post           1  2.1394e+08  213944737   15.096 0.0001022 ***  
gp             1  2.4310e+09  2431030842  171.537 < 2.2e-16 ***  
Residuals 133087 1.8861e+12  14172027  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

p4

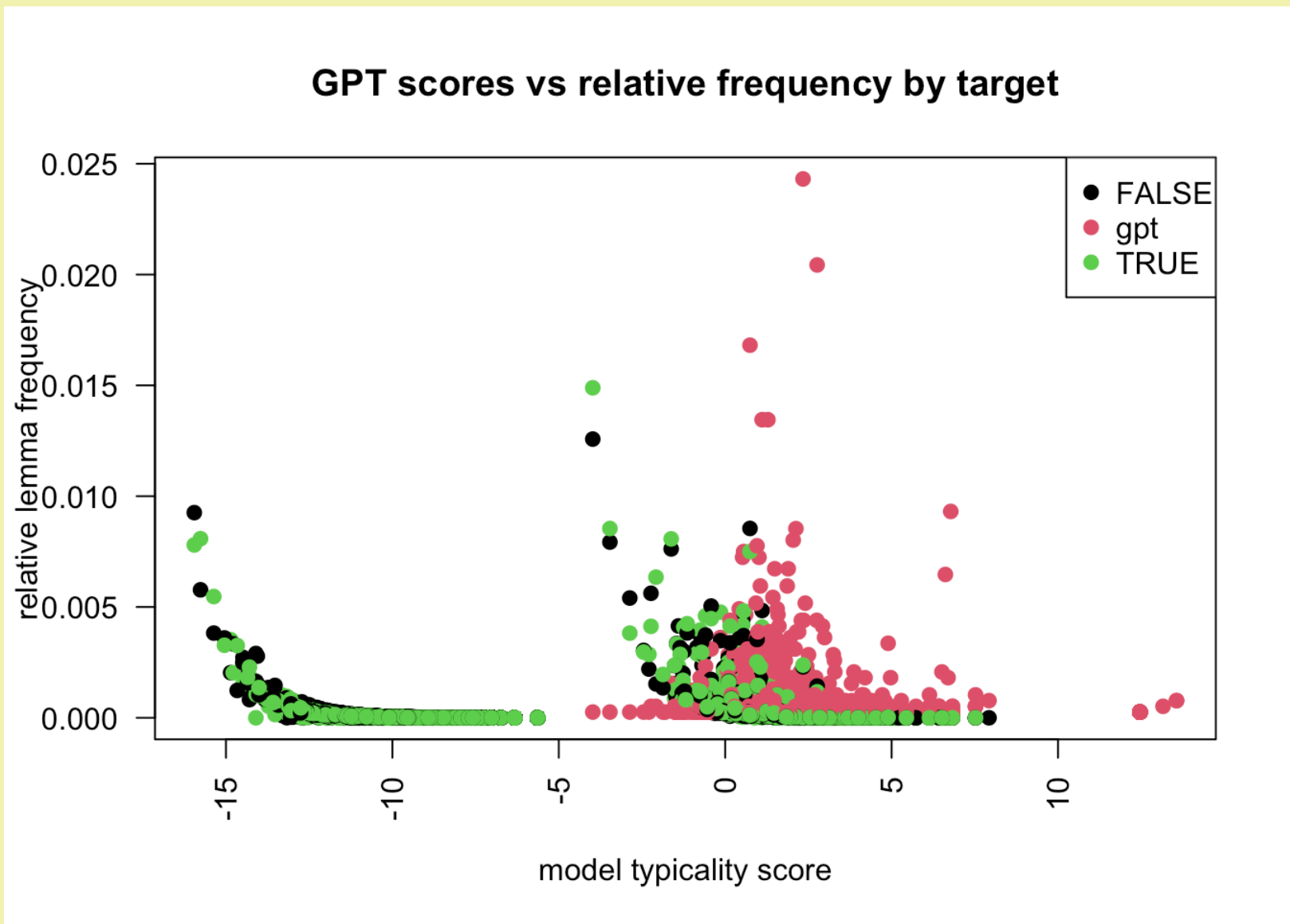


Figure 3: model typical vocab distribution

results

Yet with simple descriptive stats comparing the raw frequencies of gpt-preferred lemmas in pre- and post-gemini onset we find that in the target corpus the occurences of these lemma increase, see Figure 1. The 9 lemma where 1. the relative frequency of lemma in target=post is higher than in target=pre (n=27) and 2. that frequency exceeds the mean relative frequency is displayed in Figure 2. We observe significant frequency increases using a linear regression model (Bates et al. (2015)), cf. C3/C4, which tested the hypothesis.

c4: regression random effects model

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]  
Formula: freq_pmw ~ post + gp + (1 | lemma)  
Data: df_lg.norm  
  
REML criterion at convergence: 2490672  
  
Scaled residuals:  
    Min       1Q   Median       3Q      Max  
-48.624  -0.047  -0.032   0.008   74.360  
  
Random effects:  
Groups   Name              Variance Std.Dev.  
lemma    (Intercept) 8839356 2973  
Residual          1754704 1325  
Number of obs: 133090, groups: lemma, 99649  
  
Fixed effects:  
              Estimate Std. Error    df t value Pr(>|t|)      
(Intercept)  4.344e+02  3.966e+01 1.055e+05  10.953 <2e-16 ***  
postTRUE     1.246e+02  9.496e+00 5.602e+04  13.126 <2e-16 ***  
gp           2.932e+01  5.723e+00 1.017e+05   5.124 3e-07 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Correlation of Fixed Effects:  
      (Intr)  pTRUE  
postTRUE -0.110  
gp         0.959  0.009  
  
Type III Analysis of Variance Table with Satterthwaite's method  
              Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)      
post 302319616 302319616      1 56015 172.291 < 2.2e-16 ***  
gp   46071208 46071208      1 101716 26.256 2.996e-07 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

model explanations

We see in the model summaries that frequency increases are significant (p<0.001) when positing a random effect of lemma (model 1, cf. C4) with an estimate increase of mean frequency (pmw) per lemma by 29.32 vs. the intercept (post=FALSE i.e. target[pre]). In model 2, cf. C3 (simple model without lemma effect) we observe an estimate increase of mean frequency (pmw) of post=TRUE by 83.94 vs. the intercept. In both models gp (independent var of above mentioned gpt score: higher score > more model typical lemma) has been included as fixed effect, but vcvs. without random lemma effect that predictor lets the model naturally collapse (p<10E-16).

references

Bates, Douglas, Martin Mächler, Ben Bolker, and Steve Walker. 2015. "Fitting Linear Mixed-Effects Models Using Lme4." *Journal of Statistical Software* 67 (1): 1–48. <https://doi.org/10.18637/jss.v067.i01>.
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DIP. 2026. "DIP - Bundestagsprotokolle." Docs. *DIP - API*. Berlin. <https://dip.bundestag.de/%C3%BCber-dip/hilfe/api#content>.
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Wikipedia, and Google. 2026. "Google Gemini." *Wikipedia*. https://de.wikipedia.org/w/index.php?title=Google_Gemini&oldid=263426206.
Yakura, Hiromu, Ezequiel Lopez-Lopez, Levin Brinkmann, Ignacio Serna, Prateek Gupta, Ivan Soraperra, and Iyad Rahwan. 2025. "Empirical Evidence of Large Language Model's Influence on Human Spoken Communication." arXiv. <https://doi.org/10.48550/arXiv.2409.01754>.